

Clarification

Notation

Throughout this note,

- Q = complete compressed archive.
- $A = \{A_1, A_2, \dots, A_M\}$ = collection of all shared procedural algorithms.
- A_j = the j -th shared procedural algorithm.
- $M = |A|$ = total number of shared procedural algorithms.
- R = all reconstruction information stored in the archive.
- R_i = reconstruction information for file i .
- I_i = identifier of the shared procedural algorithm assigned to file i .
- Y_i = procedural parameters (seed) for file i .
- D_i = residual/correction data required for exact reconstruction.
- G = archive-wide information stored once (headers, model tables, checksums, metadata, version information, etc.).

Program Structure

The compressed archive is defined as

$$Q = A + R.$$

The reconstruction information is

$$R = \sum_{i=1}^N R_i + G,$$

where

$$R_i = (I_i, Y_i, D_i).$$

Thus, the decoder reconstructs each original file as

$$\text{Decode}(A_{I_i}, R_i) = F_i.$$

Optimization Objective

The objective is to minimize the total compressed archive size,

$$\min |Q| = \min \left(\sum_{j=1}^M |A_j| + \sum_{i=1}^N |R_i| + |G| \right).$$

The purpose of this objective is to determine the procedural architecture that minimizes the total archive size. It does not assume a fixed architecture beforehand. Instead, the optimizer searches for the arrangement that produces the smallest compressed archive.

Possible outcomes include:

1. A single universal procedural algorithm shared by every file.
2. Multiple shared procedural algorithms, each reused by a family of similar files.
3. Any intermediate solution between these two extremes if it produces a smaller archive.

These possibilities are represented naturally by the value of M .

1. Single shared algorithm

$$F_i = A(R_i), \quad M = 1.$$

2. Multiple shared algorithms

$$F_i = A_{I_i}(R_i), \quad M > 1.$$

Therefore, the objective function determines whether one universal procedural algorithm, multiple shared procedural algorithms, or any intermediate arrangement provides the minimum archive size rather than requiring the architecture to be chosen in advance.